

Lixin Liu (刘李鑫)

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Professional Appointments

Zhongguancun Laboratory - Assistant Researcher Jul 2026 - Present

Research Profile

My research addresses emerging architectural challenges in Low-Earth-Orbit (LEO) satellite networks as they scale to serve diverse users and support multiple operators. I have developed a coherent research agenda spanning stable routing, direct-to-cell access, capacity utilization, and service quality control, with the goal of making satellite networks more accessible, efficient, and scalable.

Education

Tsinghua University - Cyberspace Security - Doctorate Sep 2022 - Jun 2026
Tsinghua University - Cyberspace Security - Master Sep 2019 - Jun 2022
Shandong University - Software Engineering - Bachelor Sep 2015 - Jun 2019

Honors & Awards

- Jun, 2026 Outstanding Graduate, Institute for Network Sciences and Cyberspace, Tsinghua University
- Oct, 2025 Jiang Nanxiang Scholarship, Tsinghua University
- Oct, 2024 National Scholarship
- Apr, 2024 USENIX NSDI 2024 Outstanding Paper Award
- Jun, 2022 Outstanding Graduate, Institute for Network Sciences and Cyberspace, Tsinghua University

Publications

- [USENIX NSDI 2024] Lixin Liu, Yuanjie Li, Hewu Li, Jiabo Yang, Wei Liu, Jingyi Lan, Yufeng Wang, Jiarui Li, Jianping Wu, Qian Wu, Jun Liu, Zeqi Lai, Democratizing Direct-to-Cell Low Earth Orbit Satellite Networks. **Outstanding Paper Award**
- [IEEE INFOCOM 2026] Lixin Liu*, Jingyi Lan*, Bijia You, Yuanjie Li, Hewu Li, Qian Wu, Zeqi Lai, The Invisible Hand: Regaining Control of Service Quality from Outsourced Satellite Networks.
- [ACM MobiCom 2024] Yuanjie Li, Lixin Liu, Hewu Li, Wei Liu, Yimei Chen, Wei Zhao, Jianping Wu, Qian Wu, Jun Liu, Zeqi Lai, Stable Hierarchical Routing for Operational LEO Networks.
- [NINeS 2026] Lixin Liu, Jinyao Zhang, Bijia You, Yimei Chen, Jiabo Yang, Yuanjie Li, Hewu Li, Qian Wu, Zeqi Lai, Jun Liu, CrowdLink: Unlocking Idle LEO Network Capacity with User Terminals.
- [ACM HotNets 2024] Yimei Chen*, Lixin Liu*, Yuanjie Li, Hewu Li, Qian Wu, Jun Liu, Zeqi Lai, Unraveling Physical Space Limits for LEO Network Scalability.
- [IEEE/ACM IWQoS 2022] Lixin Liu, Hewu Li, Yuanjie Li, Zeqi Lai, Yangtao Deng, Yimei Chen, Wei Liu, Qian Wu, Geographic Low-Earth-Orbit Networking without QoS Bottlenecks from Infrastructure Mobility.
- [IEEE Internet Computing 2024] Lixin Liu, Wei Liu, Yuanjie Li, Hewu Li, Enabling 6G and Beyond Network Functions From Space: Challenges and Opportunities.
- [ACM HotNets 2021] Yuanjie Li, Hewu Li, Lixin Liu, Wei Liu, Jiayi Liu, Jianping Wu, Qian Wu, Jun Liu, Zeqi Lai, "Internet in Space" for Terrestrial Users via Cyber-Physical Convergence.
- [ACM MobiCom 2023] Yuanjie Li, Hewu Li, Wei Liu, Lixin Liu, Wei Zhao, Yimei Chen, Jianping Wu, Qian Wu, Jun Liu, Zeqi Lai, Han Qiu, A Networking Perspective on Starlink's Self-Driving LEO Mega-Constellation.
- [ACM SIGCOMM 2022] Yuanjie Li, Hewu Li, Wei Liu, Lixin Liu, Yimei Chen, Jianping Wu, Qian Wu, Jun Liu, Zeqi Lai, A Case for Stateless Mobile Core Network Functions in Space.

- [\[ACM SIGCOMM 2025\]](#) Yuanjie Li, Yimei Chen, Jiabo Yang, Jinyao Zhang, Bowen Sun, [Lixin Liu](#), Hewu Li, Jianping Wu, Zeqi Lai, Qian Wu, Jun Liu, Small-scale LEO Satellite Networking for Global-scale Demands.
- [\[ACM SIGCOMM 2025\]](#) Wei Liu, Yuanjie Li, Jingyi Lan, Hewu Li, Yimei Chen, [Lixin Liu](#), Jiabo Yang, Xi Long, Li Ouyang, Minghao Tang, Jianping Wu, Qian Wu, Jun Liu, Zeqi Lai, Direct-to-Cell Satellite Network without Satellite Navigation.
- [\[USENIX NSDI 2023\]](#) Zeqi Lai, Hewu Li, Yangtao Deng, Qian Wu, Jun Liu, Yuanjie Li, Jihao Li, [Lixin Liu](#), Weisen Liu, Jianping Wu, StarryNet: Empowering Researchers to Evaluate Futuristic Integrated Space and Terrestrial Networks.

Professional Service

- Reviewer, IEEE/ACM Transactions on Networking
- Reviewer, IEEE Transactions on Mobile Computing
- Reviewer, Computer Networks
- Artifact Evaluation Committee Member, ACM MobiCom 2024
- Student Volunteer, ACM SIGCOMM 2019

Community Engagement

- Guest Contributor, APNIC Blog, "**The Invisible Hand: Regaining Control of Service Quality from Outsourced Satellite Networks**", May 2026.
- Invited Speaker, Huang Danian Chaspark "Best Paper Deep Dive, Reading Top Meeting" Series, "**Democratizing Direct-to-Cell Low Earth Orbit Satellite Networks**", Nov 2025.
- Speaker, IETF 121 Satellite Networking Side Meeting, "**Stable Hierarchical Addressing and Routing for Operational Satellite Networks**", Nov 2024.
- Participant, IETF 116 Satellite Networking Side Meeting, "**Problems and Requirements of Addressing in Integrated Space-Terrestrial Network (ISTN)**", Mar 2023.